

threat-intelligence-enrichment-lab

1. Incident Overview

While reviewing a suspicious PowerShell download alert in my home SOC lab, I identified several indicators that required further investigation.

At the time of detection, it was unclear whether the activity represented legitimate administrative behavior or potentially malicious activity.

To better understand the risk, I extracted the available indicators and performed threat intelligence enrichment using multiple intelligence sources.

The objective was to gather additional context, validate the indicators, and determine whether the observed activity had been associated with known malicious campaigns.

Initial Indicators

- IP: 185.234.247.91
- Domain: update-checker[.]net
- SHA256: 44d88612fea8a8f36de82e1278abb02f6b9f8a9b1d9d8c3c4e6a8e6b6c2a5d9a

2. Threat Intelligence Findings

The indicators were enriched using multiple threat intelligence platforms, including VirusTotal, AbuseIPDB, URLHaus, and MalwareBazaar.

Each source provided additional context that helped assess the reputation, history, and potential maliciousness of the observed indicators.

The enrichment results are summarized below.

Type	Indicator	Source	Result / Score
IP	185.234.247.91	VirusTotal	Malicious
IP	185.234.247.91	AbuseIPDB	High confidence
Domain	update-checker[.]net	VirusTotal	Malicious
Domain	update-checker[.]net	URLHaus	Listed
Hash	44d88612fea8a8f36de82e...	VirusTotal	Malicious
Hash	44d88612fea8a8f36de82e...	MalwareBazaar	No matching records

Evidence Collected

To support the investigation, evidence was collected from multiple threat intelligence platforms.

Each source provided unique information regarding reputation, detections, malware associations, historical observations, and community reporting.

The evidence gathered during the enrichment process is summarized below.

Indicator	Evidence Summary	Screenshot
IP	VT detections + reputation	TI-01_VirusTotal_IP
IP	Abuse confidence score + reports	TI-02_AbuseIPDB_IP

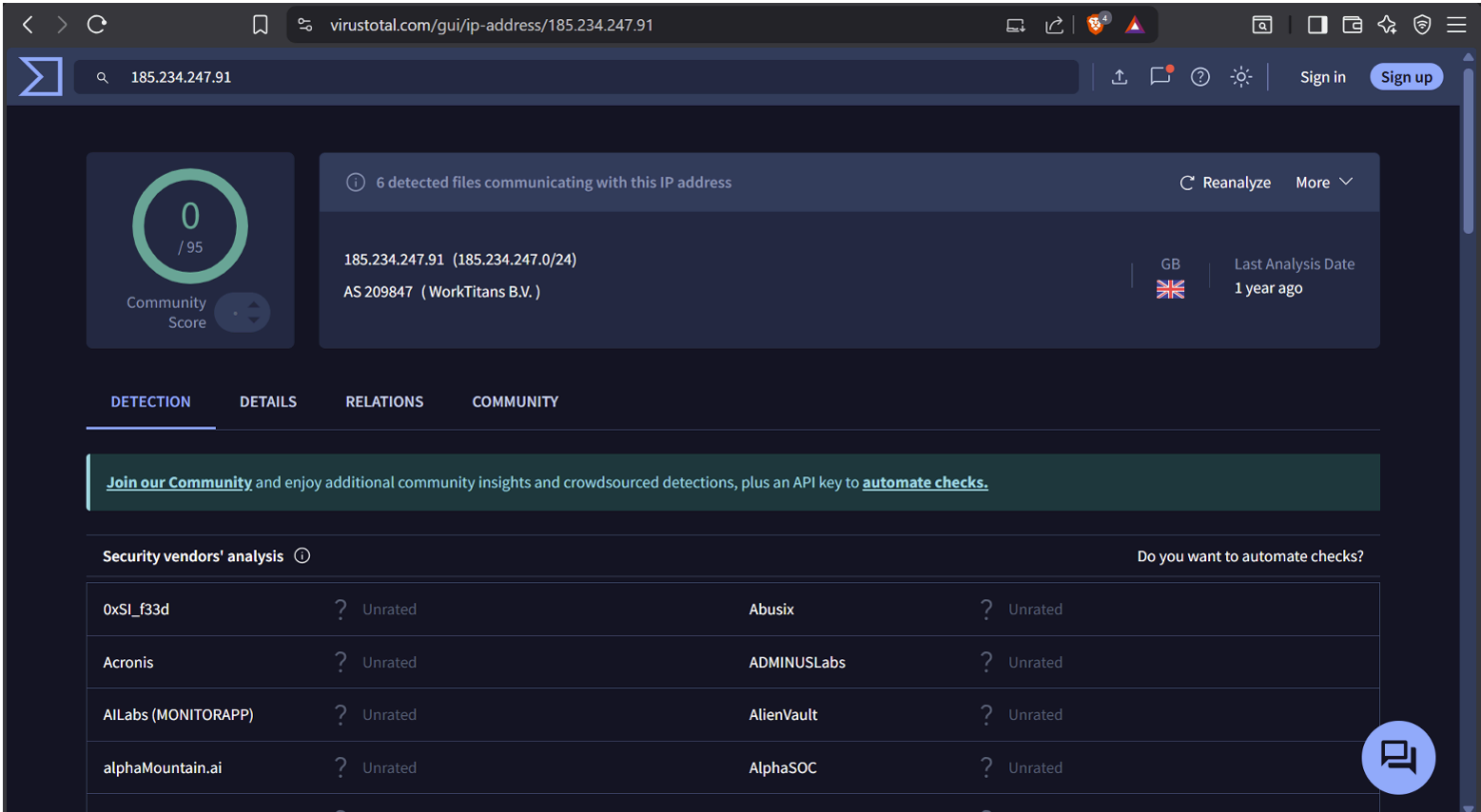
Indicator	Evidence Summary	Screenshot
Domain	VT detections + categories	TI-03_VirusTotal_Domain
Domain	URLHaus listing status	TI-04_URLHaus_Domain
Hash	VT detections + first/last seen	TI-05_VirusTotal_Hash
Hash	MalwareBazaar lookup – no matching records	TI-06_MalwareBazaar_Hash

3. Threat Intelligence Enrichment

3.1 VirusTotal (IP)

Indicator: 185.234.247.91

Source: VirusTotal



Analysis

The first step was to determine whether the IP address had been previously associated with malicious activity.

VirusTotal showed no detections from security vendors at the time of analysis (0/95 detections).

Although the absence of detections does not guarantee that an indicator is safe, the result reduced the likelihood that the IP was associated with known malicious infrastructure.

Based solely on VirusTotal results, there was insufficient evidence to classify the IP as malicious.

Analyst Assessment

At this stage of the investigation, the indicator appeared low risk.

However, additional validation was required using other threat intelligence sources before reaching a final conclusion.

Investigation Limitation

VirusTotal provides a point-in-time view and may not reflect newly registered or recently abused infrastructure.

For this reason, additional enrichment sources were consulted.

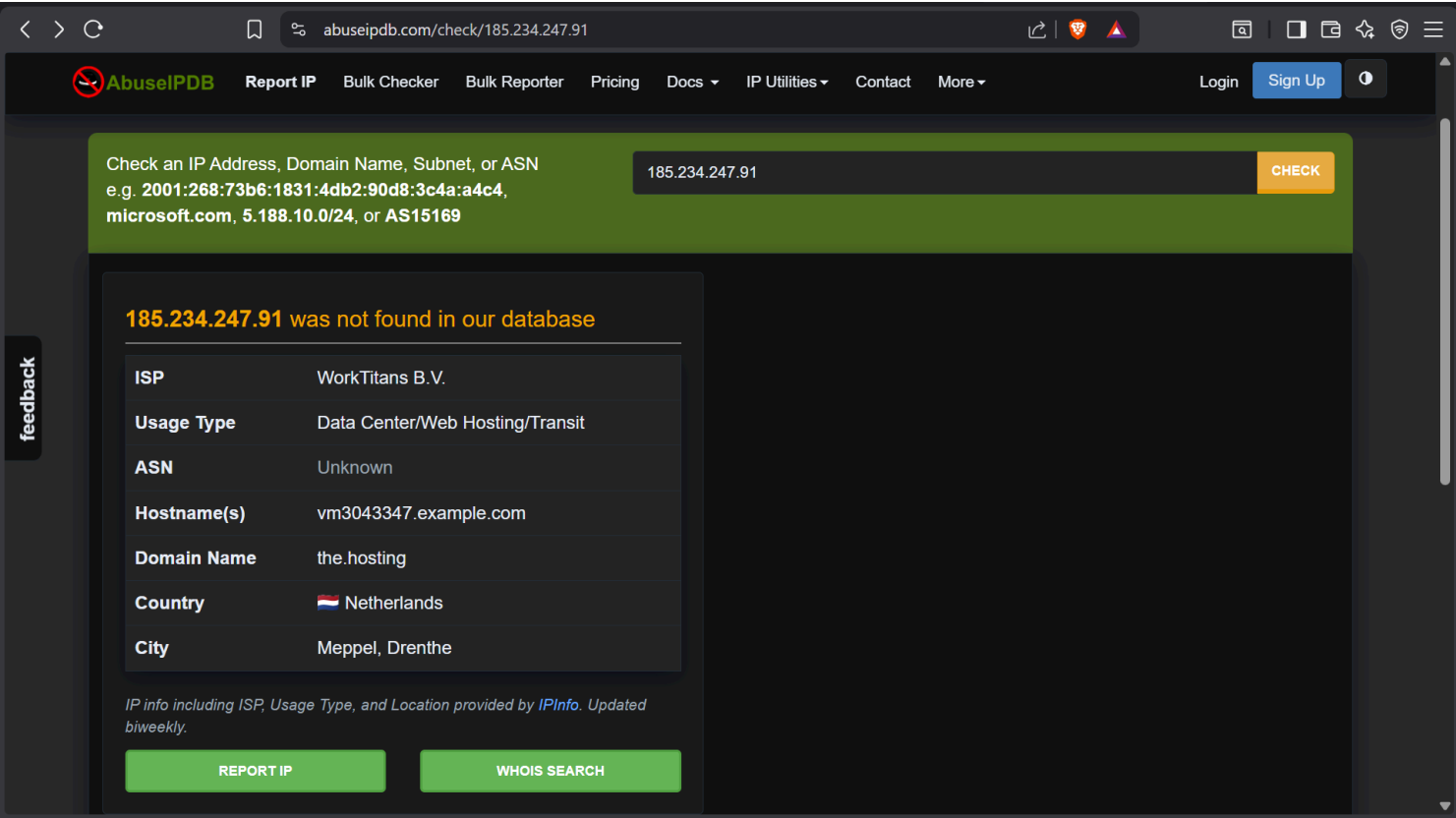
Next Step

The investigation continued with AbuseIPDB and other intelligence platforms to determine whether additional evidence existed.

3.2 AbuseIPDB (IP)

Indicator: 185.234.247.91

Source: AbuseIPDB



Analysis

The next step was to determine whether the IP address had been reported by the security community for malicious activity.

AbuseIPDB was consulted to identify any abuse reports, confidence scores, or historical observations associated with the indicator.

The search returned no reports for the IP address at the time of analysis.

While this reduced confidence that the indicator was actively associated with known malicious infrastructure, the absence of reports alone was not sufficient to classify the indicator as benign.

Analyst Assessment

At this stage of the investigation, no community-reported evidence of malicious activity was identified.

The indicator appeared low risk based on AbuseIPDB findings.

However, additional validation was required using other threat intelligence sources before reaching a final conclusion.

Investigation Limitation

AbuseIPDB relies on community reporting and may not contain newly observed or low-volume malicious infrastructure.

For this reason, the absence of reports should not be interpreted as definitive proof that the indicator is safe.

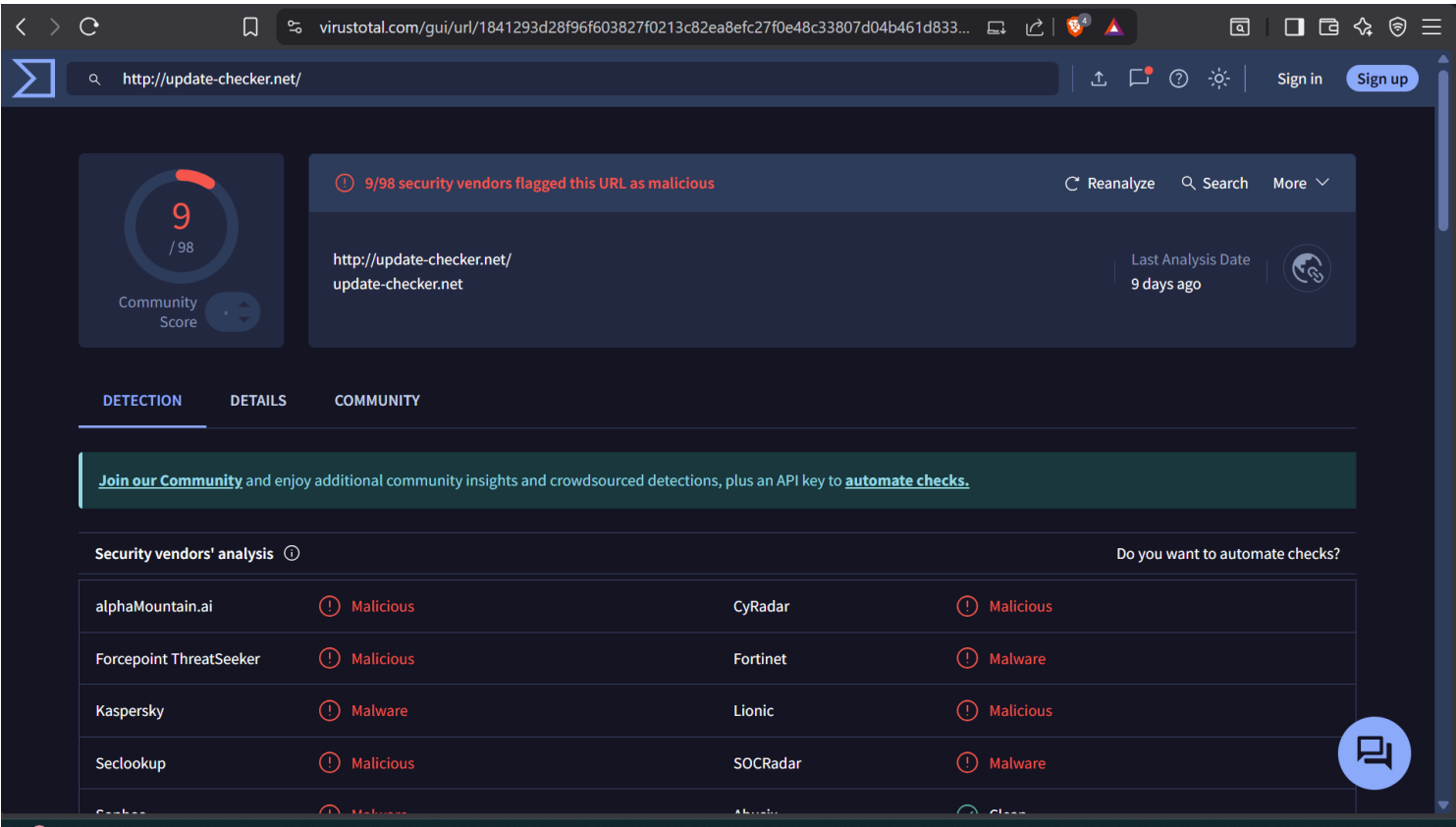
Next Step

The investigation continued by analyzing the associated domain through VirusTotal and URLHaus to determine whether additional malicious indicators existed.

3.3 VirusTotal (Domain)

Indicator: update-checker[.]net

Source: VirusTotal



Analysis

The investigation then moved to the associated domain to determine whether it had been linked to known malicious activity.

VirusTotal analysis showed that 9 out of 98 security vendors classified the domain as malicious.

Multiple detections across independent security vendors increased confidence that the domain had previously been associated with suspicious or malicious behavior.

This represented the first strong indicator that the observed PowerShell activity could be connected to known malicious infrastructure.

Analyst Assessment

Unlike the IP address, the domain showed multiple positive detections from security vendors.

The evidence significantly increased the likelihood that the observed activity was malicious rather than legitimate administrative behavior.

At this stage of the investigation, the risk level was elevated and additional validation was required.

Investigation Limitation

VirusTotal detections depend on participating security vendors and may change over time as new intelligence becomes available.

Analysts should avoid relying solely on vendor counts and continue validating indicators through additional intelligence sources.

Next Step

To further validate the domain reputation, the investigation continued with URLHaus to determine whether the domain had been associated with malware distribution or known malicious campaigns.

3.4 URLHaus (Domain)

Indicator: update-checker[.]net

Source: URLHaus

URLhaus

from ABUSE.ch

SPAMHAUS

🔍 Browse

🔔 Hunting Alerts

📂 Access Data

📄 FAQ

🏠 About

👤 Login

In order to submit a URL to URLhaus, you need to login with [your abuse.ch account](#)

Browse Database

update-checker.net

🔍 Search

🔗 URLs

⚙️ Payloads

Firstseen (UTC)	SHA256 hash	File type	Bazaar	Signature
2025-12-25 14:00:11	7b89bd1544fb05b505705d65d5761dcd32ee34217a10998dcd32faf35b5c1447	zip		
2025-12-25 14:00:08	b2e9cba1e7fdc3f3bf323ce158375c8e670a1916da06357d35b586963e0246b5	zip		
2025-12-25 13:52:13	3c4f7a674806267dd3bc30709c2c20a88f6229d273b1b31ca536a31df1e271bf	html		
2025-12-25 13:51:52	bcb05b54551930647a2a004bbc33895146c3600ba07d15dc6a7a440033371b4b	elf		
2025-12-25 13:50:47	efc58eb900f3f5181f816a20a0605accfbda5d35c346ca25f60d1b677b09d133	html		
2025-12-25 13:50:18	0fb350afc779fe3308150c5176cf33099e367807c9da8fa4adfed05f5d3f755	html		
2025-12-25 13:49:56	310df698a12c2b35c5d2c5b03dce34ec64a06202aaa315aa08e17d1a3b4da9a8	html		

Analysis

The investigation continued with URLHaus to determine whether the domain had been associated with known malware distribution activity.

URLHaus identified the domain as active malicious infrastructure and linked it to multiple payloads, including ZIP, HTML, and ELF files.

The platform also provided multiple associated SHA256 hashes, indicating that the domain had been observed distributing potentially malicious content.

These findings significantly strengthened the evidence that the domain was being used as part of a malware delivery operation.

Analyst Assessment

The URLHaus findings provided independent confirmation of the malicious reputation previously identified through VirusTotal.

The presence of multiple payloads and associated hashes increased confidence that the domain was actively involved in malware distribution.

At this stage of the investigation, the domain was assessed as high risk.

Investigation Limitation

URLHaus focuses primarily on malware distribution infrastructure and may not capture all forms of malicious activity.

Analysts should continue validating findings through additional intelligence sources and artifact analysis.

Next Step

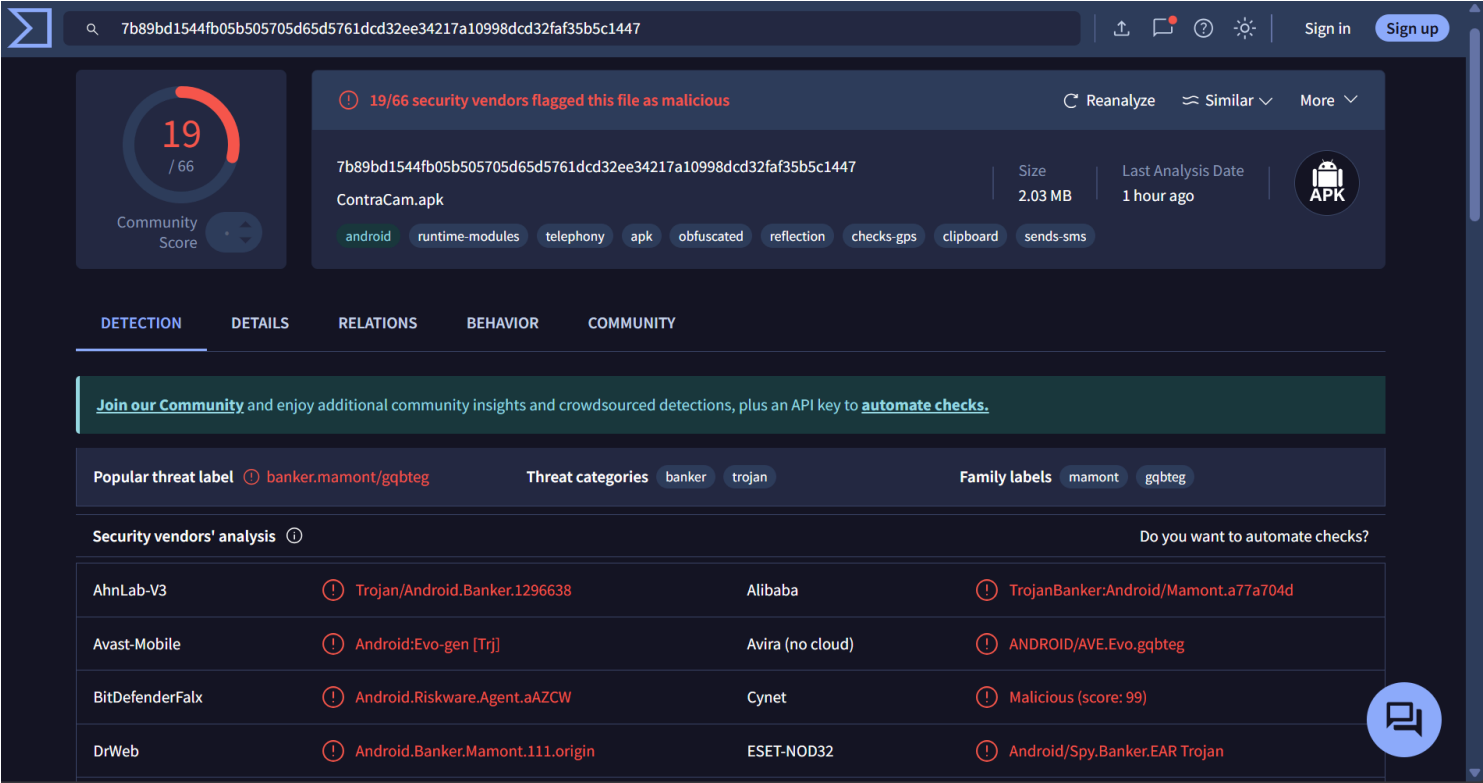
The investigation continued with file hash analysis to determine whether the downloaded artifact had previously been identified as malicious by threat intelligence platforms.

3.5 VirusTotal (File Hash)

Indicator:

SHA256: 7b89bd1544fb05b505705d65d5761dcd32ee34217a10998dcd32faf35b5c1447

Source: VirusTotal



Analysis

The investigation then focused on the downloaded file to determine whether the artifact had previously been identified as malicious.

VirusTotal showed that 19 out of 66 security vendors classified the file as malicious.

The file was identified as an Android APK named ContraCam.apk.

Several detections and labels associated the sample with Android banking trojan activity, including the Mamont malware family.

Additional tags such as obfuscated, reflection, clipboard, and sends-sms suggested behavior commonly associated with suspicious or malicious Android applications.

Analyst Assessment

At this stage of the investigation, confidence increased significantly that the downloaded artifact was malicious.

The combination of vendor detections, malware family labels, and suspicious behavioral tags supported the assessment that the file was not a legitimate application.

The risk level was assessed as high.

Investigation Limitation

VirusTotal analysis is based primarily on vendor detections, metadata, and static indicators.

Runtime behavior, persistence mechanisms, and post-execution activity were not validated within this lab environment.

Next Step

The investigation continued with MalwareBazaar to obtain additional intelligence regarding malware family classification, historical observations, and related samples.

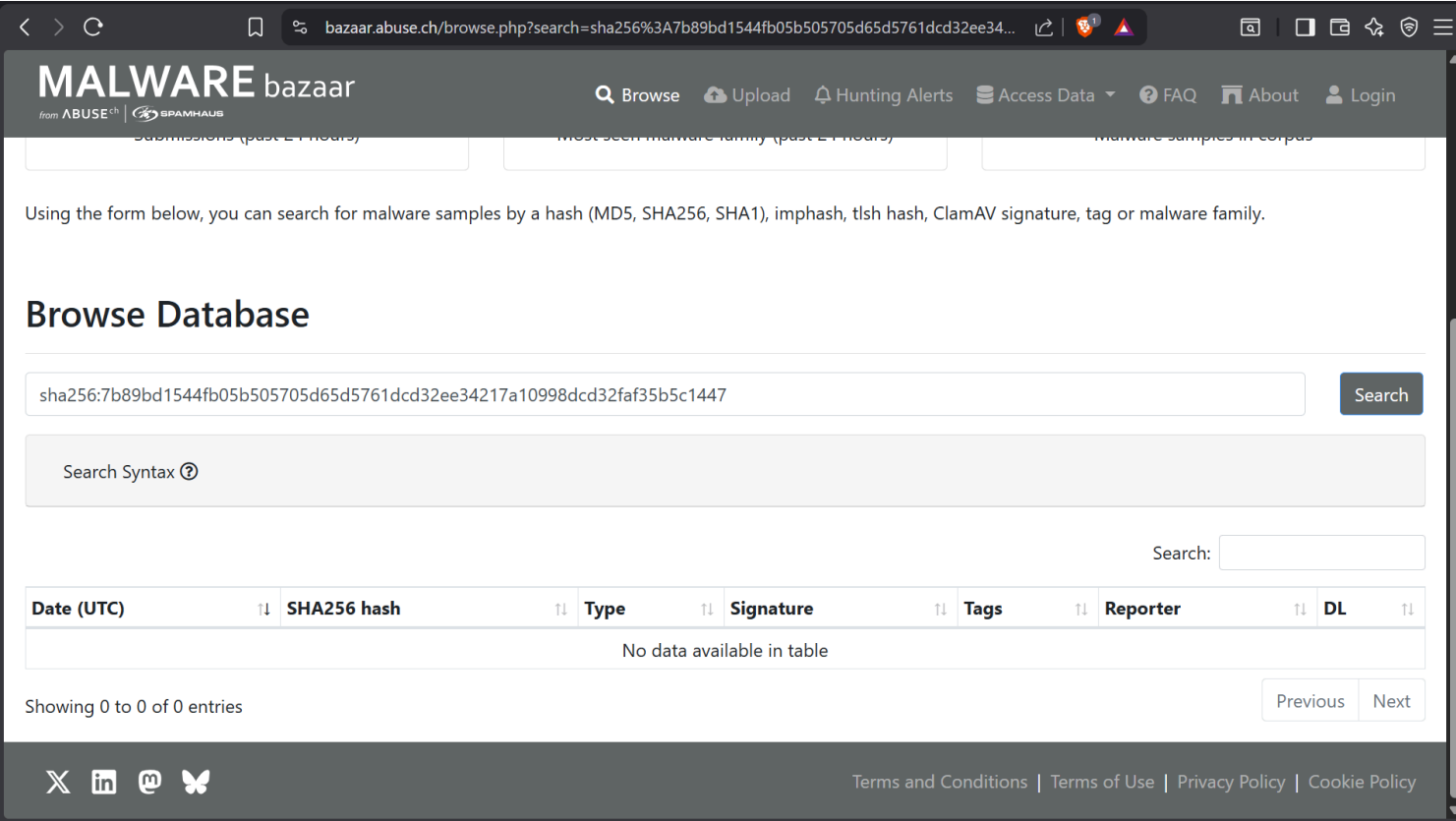
3.6 MalwareBazaar (File Hash)

Indicator:

SHA256: 7b89bd1544fb05b505705d65d5761dcd32ee34217a10998dcd32faf35b5c1447

Source:

MalwareBazaar (abuse.ch)



Analysis

The final step of the investigation was to determine whether the file hash had been catalogued within the MalwareBazaar malware repository.

A search was performed using the SHA256 hash obtained from previous intelligence sources.

At the time of analysis, MalwareBazaar returned no matching records for the indicator.

No malware family information, sample metadata, or related submissions were available.

Analyst Assessment

Unlike VirusTotal, MalwareBazaar did not provide additional intelligence regarding the file.

The absence of a MalwareBazaar record did not invalidate previous findings, as the sample had already been identified as malicious by multiple VirusTotal vendors.

The overall risk assessment remained unchanged.

Investigation Limitation

MalwareBazaar visibility depends on submitted malware samples and community contributions.

Not all malicious files are available within the repository.

Next Step

The investigation proceeded to the final correlation and overall verdict based on all collected intelligence sources.

4. Decision Log

Alert / Case:

Suspicious PowerShell Download (Lab Enrichment Case)

Date:

2025-12-25

Indicators reviewed:

- IP: 185.234.247.91
- Domain: update-checker[.]net
- File hash (SHA256): 7b89bd1544fb05b505705d65d5761dcd32ee34217a10998dcd32faf35b5c1447

Threat Intelligence Sources Used:

- VirusTotal
- AbuseIPDB
- URLHaus
- MalwareBazaar

Analysis Summary

- The IP address showed no malicious reputation across VirusTotal and AbuseIPDB.
- The domain update-checker[.]net was flagged as malicious by multiple security vendors and was listed in URLHaus as malware distribution infrastructure.
- The file hash was positively identified as malicious in VirusTotal (19/66 detections), with multiple vendors classifying the sample as an Android banking trojan associated with the Mamont malware family.
- MalwareBazaar did not return additional records for the hash at the time of analysis.
- Based on the combined intelligence sources, the observed indicators were assessed as malicious and consistent with malware delivery activity.

Final Verdict

Malicious

Confidence

High

Recommended Action

Escalate alert and investigate the affected host for evidence of malware execution, persistence, credential theft, or additional malicious activity.
